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White Matter Morphometric Changes Uniquely Predict Children's Reading Acquisition

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Abstract

This study examined whether variations in brain development between kindergarten and Grade 3 predicted individual differences in reading ability at Grade 3. Structural MRI measurements indicated that increases in the volume of two left temporo-parietal white matter clusters are unique predictors of reading outcomes above and beyond family history, socioeconomic status, and cognitive and preliteracy measures at baseline. Using diffusion MRI, we identified the left arcuate fasciculus and superior corona radiata as key fibers within the two clusters. Bias-free regression analyses using regions of interest from prior literature revealed that volume changes in temporo-parietal white matter, together with preliteracy measures, predicted 56% of the variance in reading outcomes. Our findings demonstrate the important contribution of developmental differences in areas of left dorsal white matter, often implicated in phonological processing, as a sensitive early biomarker for later reading abilities, and by extension, reading difficulties.

Keywords

brain, reading, literacy, cognitive development, childhood development, white matter

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Learning to read is critical for educational success and has an enduring influence on career opportunities and psychosocial well-being. Difficulties in learning to decode print (developmental dyslexia) are relatively common, affecting somewhere between 3% and 7% of the population (Peterson & Pennington, 2012). To explain the decoding difficulties seen in children with dyslexia, one must understand the cognitive mechanisms that are causally linked to variations in decoding skills. Robust evidence indicates that there are three main early behavioral predictors of individual differences in learning to decode in alphabetic languages: letter knowledge, phonological awareness, and rapid-naming ability (Caravolas et al., 2012; Lervåg, Bråten, & Hulme, 2009). Further, family history and socioeconomic status (SES) have been shown to be strong predictors (e.g., Bowey, 1995; Lefly & Pennington, 2000). Here, we examined whether variations in structural brain development (from kindergarten to Grade 3) are additional, sensitive, and early predictors of variations in reading ability.

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The development of phonological awareness and related skills are believed to involve temporo-parietal and caudal inferior frontal/precentral regions that are major nodes of the dorsal pathway (Hoeft et al., 2006; Hoeft, Meyler, et al., 2007; Yamada et al., 2011). Specifically, key white matter tracts, including those in left temporo-parietal regions, are important for the development of reading. For example, evidence suggests that the arcuate fasciculus (or superior longitudinal fasciculus) plays a role in language, speech-sound processing (Dick & Tremblay, 2012), word learning (López-Barroso et al., 2013), and phonological processing (Saygin et al., 2013; Thiebaut de Schotten, Cohen, Amemiya, Braga, & Dehaene, 2012; Vandermosten et al., 2012; Yeatman, Dougherty, Ben-Shachar, & Wandell, 2012). Notably, the nearby corona radiata has also been associated with reading ability (Beaulieu et al., 2005; Niogi & McCandliss, 2006; Odegard, Farris, Ring, McColl, & Black, 2009), though the underlying mechanism is less clear (Ben-Shachar, Dougherty, & Wandell, 2007).

Recently, several studies have explored whether neuroimaging measures are effective predictors of later reading abilities. These studies showed that (a) functional and structural neuroimaging measures in children were effective in predicting future reading after relevant cognitive and earlier reading skills had been accounted for (Hoeft, Ueno, et al., 2007), (b) age-dependent activation of dorsal and ventral pathways predicted reading development (McNorgan, Alvarez, Bhullar, Gayda, & Booth, 2011), (c) reading skills of children with reading disabilities but who compensated for these disabilities later were predicted by functional activation and white matter integrity (Hoeft et al., 2011), and (d) event-related potential and functional MRI measures in preliterate children correlated with reading outcomes (Bach, Richardson, Brandeis, Martin, & Brem, 2013). These studies utilized imaging data from one time point to predict reading outcomes. However, there is ample evidence suggesting that longitudinal studies measuring developmental changes are more sensitive than static measures in detecting abnormalities (e.g., Giedd & Rapoport, 2010).

Therefore, the aim of the current study was to examine whether developmental structural changes in neural circuits predict variations in reading outcomes in Grade 3. We measured changes in white matter in children, as its development appears critical for reading. We examined the degree to which white matter development uniquely predicted reading outcomes after controlling for the influence of other potentially important predictors. We hypothesized that the development of left temporo-parietal dorsal white matter, considered critical for phonological tasks, would be an early predictor of reading outcomes.

Method

Participants

As outlined in National Institutes of Health Grant K23HD054720, which was the main source of funding for the project, sample size was determined using power calculation based on our previous research and related studies that examined how neuroimaging predicts reading ability and future reading outcomes. Also as outlined in the same grant, we recruited children in 2008 and 2009, and all children who met the criteria for inclusion and had usable data were included. Thirty-eight healthy, native English-speaking children, with differing preliteracy skills and varying family histories of reading difficulty, participated in our study at ages 5 or 6 years (Time 1) and again 3 school years later (Time 2; Table 1). There were 24 males and 14 females; 33 participants were right-handed, and 5 were left-handed, as evaluated with the Edinburgh Handedness Inventory (Oldfield, 1971). These children were a subset of an original 51 participants.

Among the original 51, 6 were excluded because they dropped out or had insufficient follow-up data, and 7 were excluded because of movement during their scans at one or both time points (leaving a final sample of 38). Scores on demographic and behavioral measures at Times 1 and 2 did not statistically differ between excluded participants and included participants (all ps > .1), with the exception of those on the Time 1 Comprehensive Test of Phonological Processing (CTOPP; Wagner, Torgesen, & Rashotte, 1999) Blending Words subtest, $t(15.79) = 2.19$, $p = .044$, on which the excluded participants performed better than the included participants. Of the 38 participants, 18 met the criterion for a family history of reading difficulty on the Adult Reading History Questionnaire (ARHQ; Lefly & Pennington, 2000), using a cutoff score of greater than .4 (for at least one parent), as specified in Black et al. (2012). Participants had no neurological or psychiatric disorders, including attention-deficit/hyperactivity disorder (ADHD), were not on any medication, and had no contraindications to MRI. The Stanford University Panel on Human Subjects in Medical Research and the University of California, San Francisco Human Research Protection Program approved the study, and informed assent and consent were obtained.

Nonimaging measures

Participants completed a test battery of standardized neuropsychological assessments at both time points. Reading assessments at Time 1 included the Woodcock Reading Mastery Tests—Revised Normative Update (WRMT-R/NU; Mather & Woodcock, 1998) Letter Identification subtest, the fourth edition of the Peabody Picture Vocabulary Test (PPVT-4; Dunn & Dunn, 2007), Rapid Automatized

Table 1. Demographic Statistics of the Sample at Times 1 and 2

Measure	*M*	Range
Time 1		
Age (years)	5.52 (0.38)	5.03–6.85
Parental education (years)	16.92 (1.75)	12.50–21.00
Home Literacy Inventory	6.91 (2.01)	3.00–14.00
Adult Reading History Questionnaire (Mothers)	0.31 (0.15)	0.07–0.67
Adult Reading History Questionnaire (Fathers)	0.35 (0.13)	0.09–0.61
CTOPP Blending Words	12.03 (1.62)	8–15
CTOPP Elision	11.68 (2.75)	7–19
WRMT Letter Identification	108.76 (10.52)	80–131
Peabody Picture Vocabulary Test 4 Vocabulary	121.84 (9.87)	97–148
RAN Objects	101.53 (15.73)	57–135
RAN Colors	98.47 (15.58)	55–137
WJ-III Cognitive Battery Brief Intellectual Ability	119.61 (10.62)	92–139
Preliteracy Index 1	0 (1.00)	–2.29–1.98
Preliteracy Index 2	0 (1.00)	–2.78–2.02
CELF-4 Receptive Language Index	118.66 (11.85)	94–137
CELF-4 Expressive Language Index	117.29 (11.97)	91–144
Total gray matter volume (cc)	782.62 (68.40)	621.42–896.94
Total white matter volume (cc)	419.44 (49.95)	347.28–562.97
Time 2		
Age (years)	8.24 (0.40)	7.51–9.08
TOWRE Word Reading	111.34 (12.98)	86–138
WRMT Word Identification	116.89 (10.71)	97–139
WRMT Word Attack	114.66 (13.61)	92–146
WRMT Passage Comprehension	114.82 (9.99)	99–141
WJ-III Reading Fluency	113.26 (16.64)	91–162
WJ-III Spelling	105.92 (17.83)	74–148
TOWRE Decoding	106.66 (13.91)	77–144
RAN Objects	97.71 (16.96)	62–132
RAN Colors	97.39 (15.65)	60–123
RAN Numbers	100.21 (12.58)	76–129
RAN Letters	103.47 (11.47)	78–134
RAS Letters and Numbers	103.82 (14.19)	72–141
RAS Letters, Numbers, and Colors	101.58 (15.13)	64–143
CTOPP Blending Words	12.68 (2.29)	6–16
CTOPP Elision	10.24 (2.17)	7–16
CTOPP Memory for Digits	10.21 (2.59)	5–15
CTOPP Nonword Repetition	13.13 (2.92)	7–17
Reading-Related Index 1	0 (1.00)	–1.89–2.42
Reading-Related Index 2	0 (1.00)	–2.44–2.13
Reading-Related Index 3	0 (1.00)	–2.75–1.95
Total gray matter volume (cc)	776.02 (74.77)	552.14–931.92
Total white matter volume (cc)	445.31 (56.04)	361.84–608.23

Note: Standard deviations are given in parentheses. CTOPP = Comprehensive Test of Phonological Processing; WRMT = Woodcock Reading Mastery Tests; RAN = Rapid Automatized Naming; WJ-III = Woodcock-Johnson III Tests of Achievement; CELF-4 = Clinical Evaluation of Language Fundamentals, 4th edition; TOWRE = Test of Word Reading Efficiency; RAS = Rapid Alternative Stimulus. See the text for details of the measures.

Naming (RAN; Wolf & Denckla, 2005) Objects and Colors subtests, and phonological-awareness measures of Elision and Blending Words from the CTOPP, which were included to calculate two preliteracy indexes.

At Time 1, factor analysis using standard scores from tests related to reading identified two preliteracy indexes, which combined explained 70.3% (46.6% and 23.7%, respectively) of the total variance in prereading skills.

Letter knowledge, phonological awareness, and vocabulary measures loaded heavily on the first factor, Preliteracy Index 1. Rapid-naming measures loaded heavily on the second factor, Preliteracy Index 2.

Other measures included the Brief Intellectual Ability score from the Woodcock-Johnson III (WJ-III) Tests of Cognitive Abilities—Revised Normative Update (McGrew & Schrank, 2007), which consists of Verbal Comprehension, Concept Formation, and Visual Matching subtests and was used as a short and reliable proxy for IQ. Receptive and expressive language abilities were assessed using the fourth edition of the Clinical Evaluation of Language Fundamentals (CELF-4; Semel, Wiig, & Secord, 2003) Receptive and Expressive Language Indices. The Receptive Language Index consists of Concepts and Following Directions, Word Classes 1, Receptive, and Sentence Structure subtests, and the Expressive Language Index consists of Word Structure, Recalling Sentences, and Formulated Sentences subtests. These tests were used as indicators of early language abilities.

Measures representative of environment were SES and home literacy. At Time 1, an average of maternal and paternal years of education was used as a marker of SES because education has been demonstrated to be strongly representative of SES (e.g., Smith & Graham, 1995). The average of nonprint activities (e.g., singing songs and rhymes, watching television programs or videos) and reading activities completed at home as assessed by the Home Literacy Inventory (Marvin & Ogden, 2002) were used to measure the home reading environment.

Finally, family history of reading difficulty was assessed using the ARHQ responses from both biological parents. The ARHQ is a widely used self-report assessment that has good reliability and validity. In the current study, we treated ARHQ scores as a continuous variable and maintained separate scores for each parent to represent varying degrees of maternal and paternal risk. The correlations between the behavioral, environmental, and familial risk measures are shown in Table S1 in the Supplemental Material.

Tests at Time 2 included RAN Objects, Colors, Numbers, and Letters subtests; and the Rapid Alternative Stimulus (RAS) tests of Letters and Numbers and of Letters, Numbers, and Colors (Wolf & Denckla, 2005). Single-word reading measures included WRMT-R/NU Word Identification and Test of Word Reading Efficiency (TOWRE; Torgesen, Wagner, & Rashotte, 1999) Sight Word Efficiency subtest. Pseudoword decoding measures WRMT-R/NU Word Attack and TOWRE Phonemic Decoding Efficiency were also used. Additionally, WRMT-R/NU Passage Comprehension (a reading-comprehension measure), WJ-III Tests of Achievement (Woodcock, McGrew, & Mather, 2001) Reading Fluency subtest (a reading-fluency measure), and WJ-III Spelling (a spelling measure) were collected. Finally, Time 2 tests also included measures of phonological memory (the CTOPP Memory for Digits and Nonword Repetition subtests) and phonological awareness (CTOPP Elision and Blending Words subtests).

These reading-related standardized measures, collected at Time 2, were entered into a factor analysis. This analysis revealed three correlated factors: Reading-Related Index 1 (pseudoword decoding, single-word reading, reading fluency, spelling, and reading comprehension) explained 53.4% of the total variance in reading skills. Reading-Related Index 2 was defined by measures of rapid naming and explained 16.2% of the variance, whereas Reading-Related Index 3, which explained 6.6% of the variance, was defined by measures of phonological awareness and phonological memory. Reading-Related Index 1 was selected as the primary measure of reading outcomes for the remainder of the study, as it was most representative of reading ability and explained the most variance.

MRI data acquisition and preprocessing

Imaging data were collected at the Richard M. Lucas Center for Imaging at Stanford University in the summer and fall of 2008 and 2009 and again in 2011 and 2012. Prior to the scanning sessions, families received a packet of materials, including a CD of scanner noises and a DVD of a child going into a scanner, designed to prepare the child for the scanner sounds and environment. Children also participated in simulated MRI sessions at the center to further minimize movement (for more information, see http://cibsr.stanford.edu/participating/Simulator.html). MRI data were acquired using a GE Healthcare (Waukesha, WI) 3.0 Tesla 750 scanner 20.x software revision and an eight-channel phased-array head coil. Images acquired included an axial-oblique 3-D T1-weighted sequence with fast-spoiled gradient-recalled echo pulse sequence—inversion recovery preparation pulse (TI) = 400 ms, repetition time (TR) = 8.5 ms, echo time (TE) = 3.4 ms, flip angle = 15°, receiver bandwidth = 32 kHz, slice thickness = 1.2 mm, 128 slices, number of excitations (NEX) = 1, field of view (FOV) = 22 cm, phase FOV = 0.75, acquisition matrix = 256 × 192. Scans were inspected for the presence of excessive motion by three experienced imaging researchers blind to the purpose of the study, and scans were excluded when agreed on by the researchers.

In addition, at Time 2, high-angular resolution diffusion-imaging (HARDI) scans were collected in a subset of the children to examine the likelihood that the results from T1 structural MRI are spatially proximal to a certain white matter tract. A total of 150 diffusion-encoding gradient directions with the diffusion sensitivity of b = 2,500 s/mm^2 were sampled using the following parameters: TR = 5,000 ms, TE = 82.6 ms, matrix size = 128 × 128, spatial in-plane resolution = 2.0 × 2.0 mm, slice thickness = 3.0 mm.

The Statistical Parametric Mapping 8 (SPM8) statistical package (Wellcome Department of Cognitive Neurology, Institute of Neurology, London, England), including the Voxel-Based Morphometry and Diffeomorphic Anatomical Registration Through Exponentiated Lie Algebra toolboxes (Ashburner, 2007), was used to preprocess the structural MRI images. Images were bias-field corrected and segmented into gray matter, white matter, and cerebro-spinal fluid. Nonlinear registration was used to normalize segmented images to Montreal Neurological Institute (MNI) space, and these images were then modulated. Each participant's data were visually inspected to ensure it was usable. An 8-mm full-width half-maximum isotropic Gaussian kernel was applied for smoothing.

HARDI images were first visually inspected for possible artifacts. All additional preprocessing was conducted in ExploreDTI (Leemans & Jones, 2009). Coregistration, Adjustment, and Tensor-Solving–a Nicely Automated Program (CATNAP) was used to correct eddy current and motion-induced artifacts with the accepted reorientation of the *b*-matrix (Leemans & Jones, 2009). No participants were excluded because of motion in HARDI scans. The root-mean-square of estimated translational movement was 1.23 mm, with a standard deviation of 0.73 mm. Of note, 6 subjects had a motion greater than 1.5 mm. A diffusion tensor model was estimated using nonlinear least-square fitting. A whole-brain tractography was constructed for each participant's data using a seed point [2 2 2] with additional parameters of an angle threshold of 40° and a fiber length range of 50 to 500 mm.

Statistical significance threshold

The statistical significance threshold for whole-brain analyses was determined by Monte Carlo simulations using 3dClustSim in AFNI (Cox, 1996). All reported clusters were significant at p = .05, corrected for multiple comparisons, which dictated that results were limited to a voxel height of p < .005 and a cluster size of at least 257 contiguous voxels by volume. All reported coordinates are in MNI space.

Whole-brain analyses of white matter volume

To examine the relationship between white matter development and reading outcomes, we computed Reading-Related Index 1, a composite reading measure based on factor analysis of reading measures from Time 2. White matter development was measured by the difference in volume between Time 1 and Time 2 images. Further, this basic analysis was repeated to examine the effect of white matter development after controlling for other factors commonly associated with reading outcomes, including Time 1 measures of preliteracy (Preliteracy Indexes 1 and 2, determined by factor analysis of Time 1 preliteracy measures), family history (maternal and paternal), environmental measures (home literacy and SES), and cognitive and linguistic capacities (measures of IQ and receptive and expressive language). Similar analyses were repeated for gray matter and Time 1 gray and white matter but are not reported as no regions showed specific effects predicting reading outcomes above and beyond other measures, such as environment, cognition, preliteracy, and family history at thresholds corrected for multiple comparisons. We analyzed the relationship between change in white matter volume and reading outcomes by regressing out each nuisance variable (i.e., environment, preliteracy, family history, cognitive capacities) separately. This was done so that we could observe the influence of each nuisance variable independently. However, we also examined the relationship between change in white matter volume and reading outcomes when all nuisance variables were entered into one model. These results can be found in Figure S1 in the Supplemental Material.

Analyses based on a priori regions of interest (ROIs) from the past literature

We next investigated how brain-imaging measures fared against measures commonly used to identify risk for developing reading disabilities or poor reading. Stepwise multiple regression analysis was performed with Reading-Related Index 1 as the dependent variable and several behavioral and imaging independent variables. Behavioral independent variables included Preliteracy Indexes 1 and 2, maternal and paternal history, home literacy environment, SES, IQ, and receptive and expressive language, all acquired at Time 1. Neuroimaging independent variables included change in volumes from Time 1 to Time 2 in white matter ROIs identified from past diffusion-imaging literature that reported coordinates of locations that had shown significant association with reading in children (Table 2). In the second stepwise regression analysis, we entered all ROIs into one model in order to determine how much of the variance was explained by developmental neuroimaging measures (i.e., Time 2 minus Time 1) alone. In the final stepwise regression analysis, Time 1 white matter volumes of the eight ROIs (A–H) listed in Table 2 were entered in place of developmental measures (i.e., change in white matter volume from Time 1 to 2).

In this analysis, ROIs were defined as spheres with a 5-mm radius from the peak coordinates of white matter

Table 2. Results of A Priori Regions-of-Interest (ROI) Analyses Based on Previous Voxel-Based Studies

	Peak MNI coordinates in present study					MNI coordinates in previous study		
ROI	*x*	*y*	*z*	Location	Previous study	*x*	*y*	*z*
A	–29	–42	28	Left temporo-parietal white matter	Deutsch et al., 2005	–28	–24	27
					Niogi & McCandliss, 2006	–28	–30	22
				Posterior limb of the internal capsule	Beaulieu et al., 2005	–32	–18	22
						–28	–11	24
						–28	–14	24
B	–42	–54	28	Left temporo-parietal white matter near arcuate fasciculus	Saygin et al., 2013	–42	–54	28
C	–14	–9	42	Left superior corona radiata	Odegard, Farris, Ring, McColl, & Black, 2009	–14	–9	42
D	–11	24	37	Left anterior centrum semiovale	Keller & Just, 2009	–10	20	38
						–12	28	36
E	28	–42	28	Right temporo-parietal white matter	Deutsch et al., 2005	28	–42	28
F	–21	0	27	Left superior corona radiata (inferior to ROI C)	Odegard et al., 2009	–21	0	27
G	–4	–33	19	Left posterior corpus callosum	Odegard et al., 2009	–4	–33	19
H	–27	–29	5	Left temporal white matter	Nagy, Westerberg, & Klingberg, 2004	–27	–29	5

Note: MNI = Montreal Neurological Institute.

regions reported in previous literature as being important for reading (Beaulieu et al., 2005; Deutsch et al., 2005; Keller & Just, 2009; Nagy, Westerberg, & Klingberg, 2004; Niogi & McCandliss, 2006; Odegard et al., 2009; Saygin et al., 2013). ROIs were combined if their spatial locations overlapped (Table 2). Studies with coordinates focusing on samples of English-speaking children were included. Because most studies of ventral pathways were in native space, there is a noted bias toward coordinates of the dorsal rather than ventral pathway. For example, studies such as those of Vandermosten et al. (2012) and Yeatman et al. (2012) were excluded even though these researchers found a significant effect of reading in the ventral pathway. Mean white matter volumes of the eight ROIs were extracted for each participant at each time point, and volume differences were calculated.

Fiber tracking

Following the procedure used in Vandermosten et al. (2012), we examined white matter fibers in the participants from whom we collected HARDI images at Time 2 (n = 28). Tractography was performed in native space. The two clusters from analyses of white matter development that predict reading above and beyond preliteracy, cognitive, environmental, and family-history measures were used as two independent seed ROIs on each subject's HARDI images. Because of the proximity of our clusters to left temporo-parietal white matter, we suspected the clusters to contain the arcuate fasciculus and potentially the corona radiata. To confirm this, we performed fiber tracking of the arcuate fasciculus and corona radiata in 5 subjects (see the following paragraph for details). Once confirmed, resultant tracts from our whole-brain cluster seeds were assessed visually in each participant to determine which tract was most probable in accounting for volumetric changes.

To confirm the presence of the corona radiata and arcuate fasciculus, we performed atlas-based definition for each of the 5 subjects in native space by manually placing ROIs, on the basis of prior anatomical knowledge of the white matter tracts' trajectory. Seed ROIs and AND ROIs represent brain regions that all fibers of the tract must pass through, whereas NOT ROIs exclude fibers running through this region. Figure S2 in the Supplemental Material depicts where the different ROIs (seed, AND, NOT) were placed for the corona radiata and the three segments of the arcuate fasciculus. To delineate the segments of the arcuate fasciculus (Catani, Jones, & ffytche, 2005), we had to adapt the protocol of Wakana et al. (2007) because it provides ROIs to delineate the long segment but not to separate the anterior and posterior segment of the arcuate fasciculus (for a similar approach, see Vandermosten et al., 2012). With regard to tracking the corona radiata, there is no generally accepted practice, as the corona radiata is a large fiber bundle, which spans the cortical surface, basal ganglia, and thalamus. First, we selected an ROI on the basis of the study of Niogi & McCandliss (2006); this ROI included the blue voxels on the diffusion tensor model at an axial slice at the level of

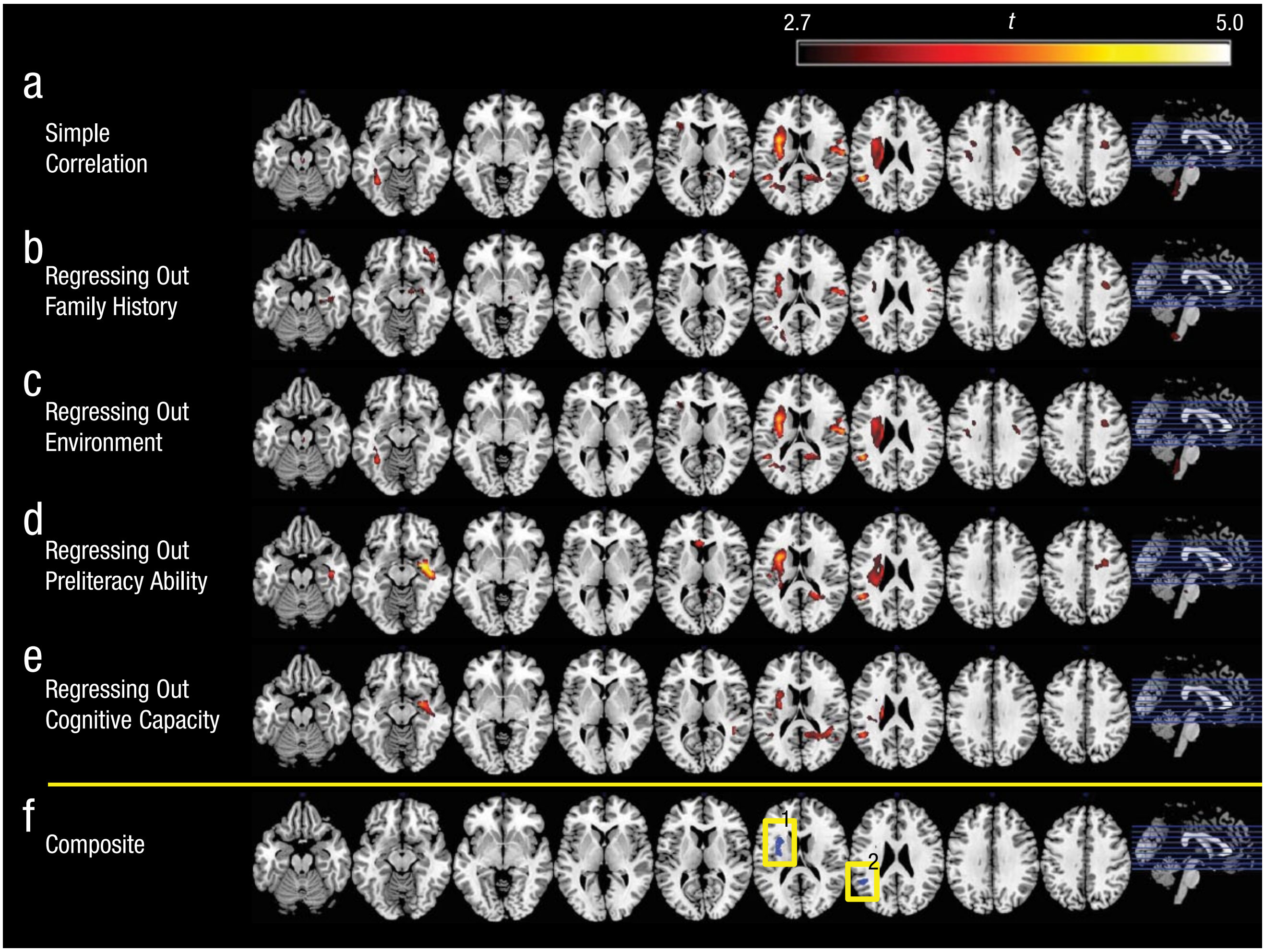

Fig. 1. White matter regions whose increase in volume from Time 1 to Time 2 was correlated with a composite measure of reading at Time 2. The first row (a) shows results from a simple correlational analysis. The second through fifth rows show partial correlations after controlling for (b) family history, which included maternal and paternal Adult Reading History Questionnaire (Lefly & Pennington, 2000) scores; (c) environment, which included Home Literacy Inventory (Marvin & Ogden, 2002) exposure to print and nonprint activities and average maternal and paternal years of education; (d) preliteracy ability, which included two composite measures of commonly used preliteracy indicators, including measures of phonological awareness, letter knowledge, rapid naming, and vocabulary; and (e) cognitive capacity, which included measures of IQ and expressive and receptive language. The regions of overlap from these correlational analyses are highlighted by the yellow boxes in (f). These regions correspond to Regions 1 and 2 in Table S2. The level of significance used was p = .05, corrected. The images at the right show the locations from which the depicted slices were taken. Results showing the relationship between change in white matter volume and reading outcomes when all nuisance variables were entered into one model are reported in Figure S1. The left hemisphere is pictured on the left side of each brain image.

the body of the corpus callosum. This ROI was used as a seed region for fiber tracking (see Fig. S2). Next, interhemispheric fibers or fibers that belonged to the arcuate fasciculus were excluded by placing a NOT ROI.

Results

White matter development and reading outcomes

Whole-brain analysis of voxel-by-voxel white matter volume yielded a significant positive correlation between Reading-Related Index 1 and increase in regional white matter volume between Time 1 and 2 in a number of white matter regions encompassing bilateral dorsal fronto-parieto-temporal and left ventral occipito-temporal regions (Fig. 1; Table S2 in the Supplemental Material; $p < .05$, corrected).

To control for family-history, environmental, and behavioral variables, we ran a series of whole-brain partial-correlation analyses. Nuisance variables included family history, environment, preliteracy, and cognition and early-language constructs (Figs. 1b–1e; Table S2). We regressed out each of these key constructs in

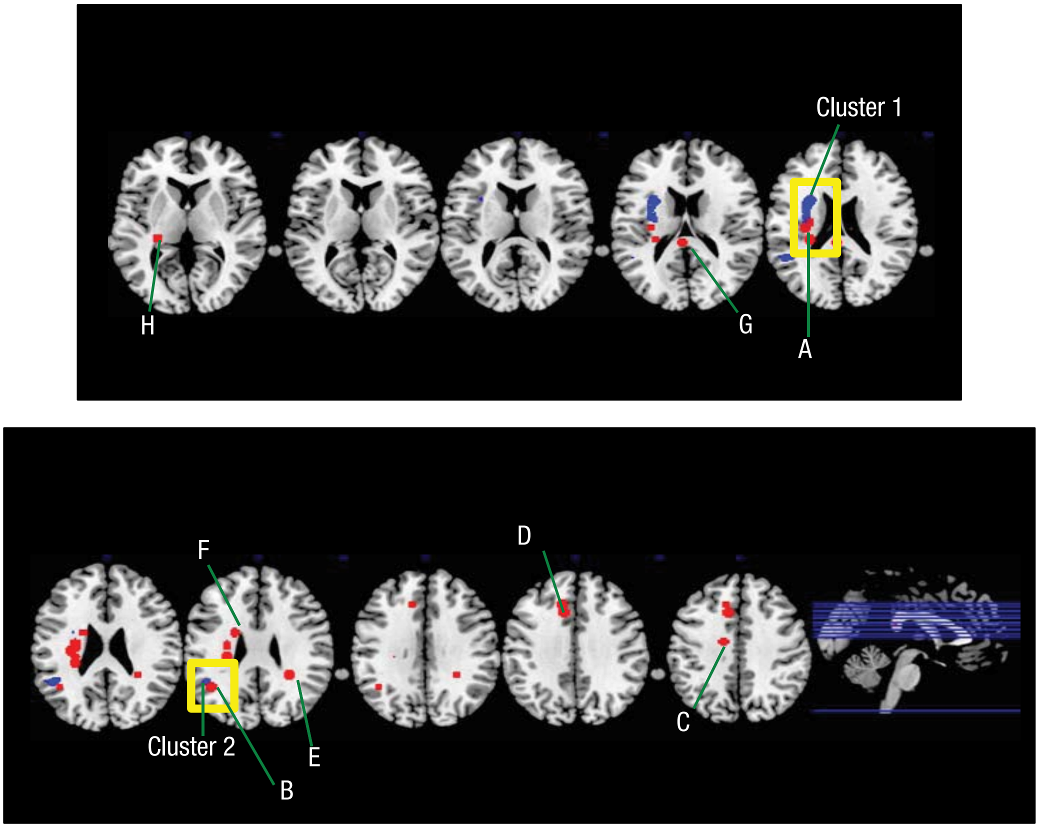

Fig. 2. Results for a priori regions of interest (ROIs) based on previous voxel-based studies. Axial views of overlays of selected ROIs (labeled according to Table 2) are shown in red, and volumetric regions that predict reading above and beyond family history, environment, preliteracy ability, and cognitive factors are shown in blue (see Fig. 1f). The yellow boxes highlight regions where there is overlap of the two in the left temporo-parietal region. The image at the bottom right shows the locations from which the depicted slices were taken. ROIs from ventral pathways were not included because the majority of past studies that have reported the ventral pathway have performed tractography or analyses in native space. The left hemisphere is pictured on the left side of each brain image.

independent generalized linear models to investigate the influence of each factor separately. A number of white matter regions showed significant positive correlations with white matter volume and a composite measure of reading outcomes in both left and right white matter, including many that overlapped with results from simple correlation analysis (Table S2).

Conjunction analyses, including all key control variables, revealed two left temporo-parietal regions that were significantly and positively associated with increases in white matter volume and reading outcomes. In other words, left temporo-parietal white matter development predicted reading outcomes above and beyond familial risk, environment, preliteracy ability, and cognitive capacity (Fig. 1f, Table S2). The larger left temporo-parietal white matter region (Cluster 1; 2,633 voxels) overlapped with the most highly replicated findings from previous diffusion-imaging studies of reading (Figs. 1f and 2; ROI A). A second region (Cluster 2; 1,037 voxels) was proximal to the supramarginal and angular gyri and overlapped with a region reported recently in Saygin et al. (2013; Figs. 1f and 2; ROI B). Figure S1 presents clusters that survived when all nuisance variables were entered into one model. These clusters included two left temporo-parietal clusters as well as clusters in the right hemisphere. Finally, Figure 3 displays the relationship between reading outcomes and change in white matter volume. The figure demonstrates that there is considerable variability in changes of white matter volume over time, with some participants increasing, some remaining stable, and some even decreasing.

A priori ROI-based approaches

Using multiple regression analyses, we examined whether developmental changes in white matter volume of brain regions identified in previous diffusion-imaging studies as being associated with reading would be as predictive as nonimaging measures that are known to predict reading outcomes. ROIs identified in previous studies

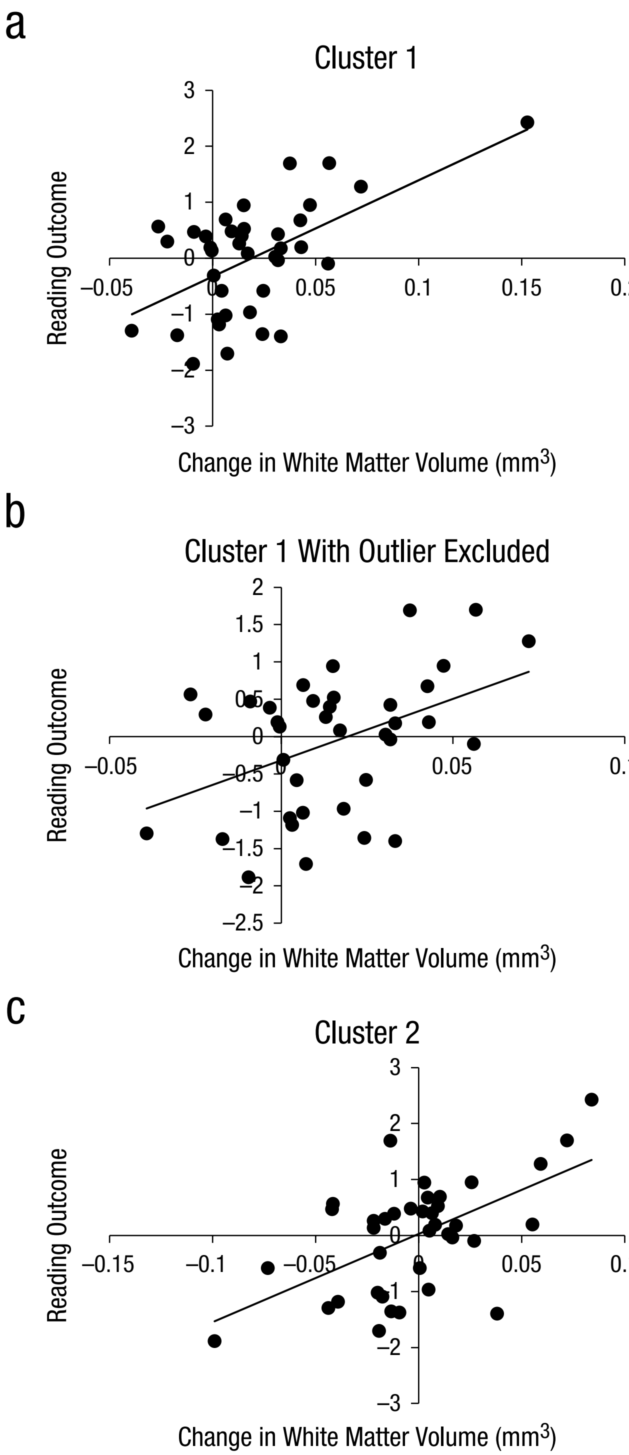

Fig. 3. Scatter plots (with best-fitting regression lines) showing the relationship between development of white matter (change in volume between Time 2 and Time 1) and Time 2 reading outcomes (Reading-Related Index 1; see the text for details). The plot in (a) shows Cluster 1 with all participants included ($N = 38$). The plot in (b) shows Cluster 1 with an outlier omitted ($n = 37$). This individual was not considered an outlier for any other measures (demographic, behavioral, Time 1 or Time 2 white matter volume in Cluster 1, and change in volume in other clusters) and thus was included in the remaining analyses. The plot in (c) shows Cluster 2 with all participants included ($N = 38$). The participant omitted in (b) was not an outlier for Cluster 2.

(summarized in Table 2; Beaulieu et al., 2005; Deutsch et al., 2005; Keller & Just, 2009; Nagy et al., 2004; Niogi & McCandliss, 2006; Odegard et al., 2009; Saygin et al., 2013) and measures of preliteracy (Preliteracy Indexes 1 and 2), environment, cognition, and family history were entered into the stepwise regression model. Explanatory variables that were retained in the final model included Preliteracy Index 1, which explained 34.2% of the variance, $F(1, 36) = 20.20$, $p < .001$, ROI A (left temporo-parietal white matter and posterior limb of the internal capsule; see Table 2), $F(1, 35) = 11.66$, $p = .002$, and ROI C (left superior corona radiata; Table 2), $F(1, 34) = 6.21$, $p = .018$, both regions located in left temporo-parietal white matter, which explained an additional 21.6% of the variance (Table 3). The inclusion of the remaining candidate variables did not increase the variance accounted for in the model including ROI B, which overlapped with Cluster 2 in the whole-brain analysis. Variables were retained on the forward step at an R^2 change of $p < .05$ and removed on backward step at an R^2 change of $p > .1$. In the second analysis, when the neuroimaging developmental measures were entered into a model without the behavioral measures, ROI A accounted for 20.6% of the variance in reading outcomes and, combined with ROI C, explained 27.4% of the variance. The remaining ROIs were not significant predictors.

In the final ROI-based regression analysis, in which Time 1 white matter volume of ROIs listed in Table 2 were entered instead of developmental measures (i.e., Time 2 minus Time 1), the neuroimaging measures were not statistically significant predictors and so were not included in the final model. When Time 1 neuroimaging measures were used, instead, a model which included Preliteracy Index 1, $F(1, 36) = 20.20$, $p < .001$, CELF-4 Expressive Language, $F(1, 35) = 5.27$, $p = .028$, and maternal family history, $F(1, 35) = 7.01$, $p = .012$, provided the best prediction of reading outcomes, explaining 50% of the variance.

Diffusion-imaging correlates

At Time 2, HARDI scans in a subset of the children were used to examine the likelihood that the results from T1 structural MRIs were spatially proximal to a particular white matter tract. The fibers running through the larger cluster (Cluster 1) were difficult to determine because of the proximity of this region to many crossing fibers. Nevertheless, in all 28 subjects, superior-inferior fibers were observed, which denotes the superior corona radiata in this region. In 24 of the subjects, fibers of the long segment of the arcuate fasciculus ran through this cluster-based seed.[1] In 11 subjects, there were some fibers of the anterior segment of the arcuate fasciculus. Cluster 2 revealed a presence in 27 of the 28 subjects of fibers of the posterior segment of the arcuate fasciculus (in

Table 3. Results of the Stepwise Multiple Regression Predicting Reading Outcomes at Time 2

Model and predictor	β	*t*	*p*
Model 1 (R = .599, adjusted R^2 = .342)			
Preliteracy Index 1	0.599	$t(36) = 4.494$	< .001
Model 2 (R = .721, adjusted ΔR^2 = .150)			
Preliteracy Index 1	0.545	$t(35) = 4.613$	< .001
ROI A	0.404	$t(35) = 3.415$	.002
Model 3 (R = .770, adjusted ΔR^2 = .066)			
Preliteracy Index 1	0.535	$t(34) = 4.846$	< .001
ROI A	0.817	$t(34) = 4.101$	< .001
ROI C	–0.494	$t(34) = –2.491$	.018

Note: Preliteracy Index 1 was a composite of measures of phonological awareness, letter knowledge, and vocabulary. ROI = region of interest.

1 subject, no fibers were found). Additionally, in 10 of 28 of the subjects, the anterior or long segment of the arcuate fasciculus was present (Fig. 4).

Discussion

We have presented evidence that developmental increases in left dorsal white matter volume uniquely contribute to the prediction of reading outcomes at a time when children have typically become proficient readers. Two left temporo-parietal white matter regions predicted reading outcomes after we controlled for a range of predictors generally considered important for literacy development (Clusters 1 and 2; Fig. 1f). These predictors included maternal and paternal familial history of reading difficulty, SES, home-literacy environment, general linguistic and

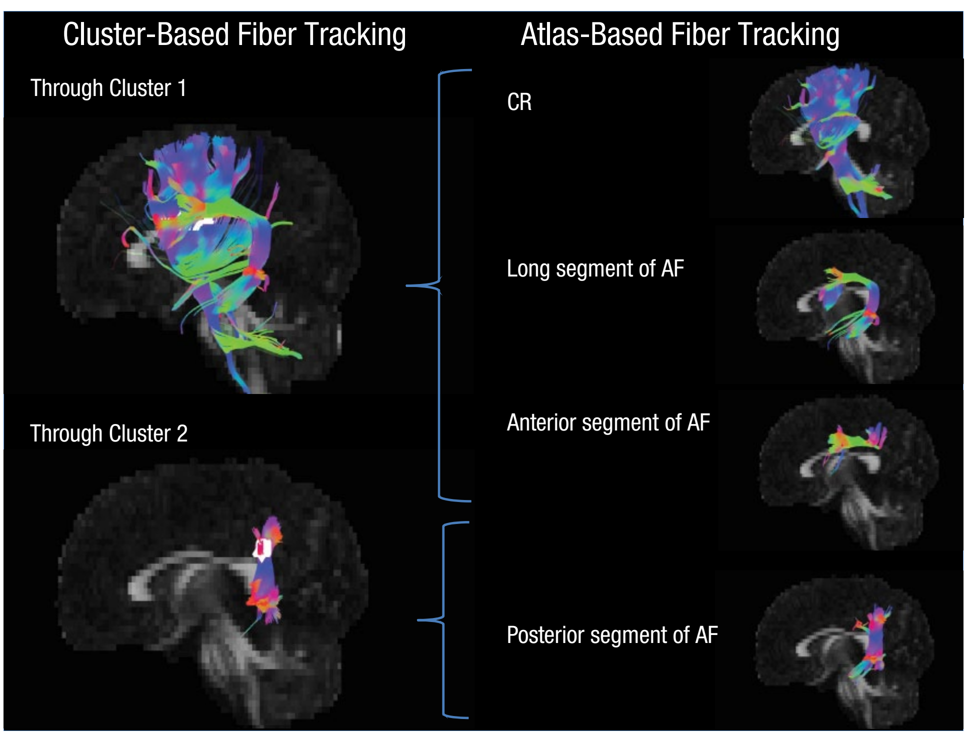

Fig. 4. Seed tracking for a representative subject for whom cluster-based and atlas-based fibers are delineated. For the cluster-based results, the upper image shows the fibers running through Cluster 1, and the lower image shows the fibers running through Cluster 2 (both clusters are shown in white). To determine which anatomically validated white matter tracts these cluster-based fibers belonged to, we delineated for this subject the corona radiata (CR) and arcuate fasciculus (AF), split up into its three segments on the basis of an atlas-based protocol (see the Method and Fig. S2 for more details).

cognitive capacities, and preliteracy measures, such as phonological awareness, letter knowledge, vocabulary, and rapid naming. The larger cluster identified as critical for the prediction of reading development in this study is the most replicated white matter location associated with reading from previous cross-sectional neuroimaging studies (Fig. 2; i.e., Beaulieu et al., 2005; Deutsch et al., 2005; Klingberg et al., 2000; Nagy et al., 2004; Niogi & McCandliss, 2006). In a complementary ROI-based analysis in which ROIs were derived from previous reading-related diffusion-imaging studies, increases in volume in two ROIs located in left dorsal white matter (ROI A overlapping with Cluster 1, and ROI C) explained 21.6% of unique variance in addition to kindergarten preliteracy abilities, which on its own explained 34.2% of the variance of reading outcomes at Grade 3 (Table 3).

Phonological processing—in particular, phonological awareness—is crucial to the development of reading skills, enabling beginning readers to associate basic component sounds of words with visual representations (Liberman, Shankweiler, Fischer, & Carter, 1974). The development of phonological-awareness skills is believed to involve the dorsal pathway that connects temporo-parietal regions with caudal inferior frontal and precentral regions (Hoeft et al., 2006; Hoeft, Meyler, et al., 2007; Yamada et al., 2011) and is also believed to be heavily involved in earlier stages of reading acquisition (Yamada et al., 2011). More specifically, the arcuate fasciculus is considered crucial for phonological skills (Thiebaut de Schotten et al., 2012; Vandermosten et al., 2012; Yeatman et al., 2012), as well as other skills important to literacy acquisition, such as speech-sound processing (Dick & Tremblay, 2012) and word learning (López-Barroso et al., 2013). Whereas most of these neuroimaging studies have employed a cross-sectional design, a recent study by Yeatman et al. (2012) demonstrated that the rate at which fractional anisotropy (a measure calculated in diffusion imaging that reflects the degree of diffusion anisotropy of water through biological tissue; Pierpaoli & Basser, 1996) changed over time in both the arcuate fasciculus and inferior longitudinal fasciculus predicted reading outcomes in an accelerated longitudinal design. Our findings complement and add to the understanding of the development of white matter and reading abilities by utilizing a different imaging modality (T1-weighted MRI) and a longitudinal design beginning at a younger age (~4 years on average younger) at the start of formal reading instruction.

One possible interpretation of our data is that structural brain differences causally influence early variations in reading development. However, we measured *changes* in the volume of critical left-hemisphere fiber tracts during a time when children were making large strides in learning to read. Changes in volume in the structures we identified may in part reflect changes in the amount of myelin present (Fjell et al., 2008). The degree of myelination is related to the electrical activity of a particular axon (Ishibashi et al., 2006) and may reflect differences in experience that drive differences in brain development. Thus, changes in volume in the left-hemisphere fiber tracts that relate to reading acquisition in the same time window may reflect differences in brain growth that are at least partly the product of experiential influences. There may, however, be genetic influences that constrain the degree of plasticity in these brain areas because of different environmental influences. Further studies are required to examine these complex issues of cause and effect. It would be interesting, for instance, to examine in a randomized trial the extent to which intensive reading practice is associated with changes in the volume of brain areas identified in this study.

Using diffusion MRI data from Time 2, we identified the probable fibers underlying the clusters identified in whole-brain analyses. In Cluster 1, the mixed probability of the volumetric differences residing in the arcuate fasciculus and the superior corona radiata reflect the difficulty in differentiating fiber tracts in the left temporo-parietal region of white matter because of close proximity. In almost all subjects, our larger cluster contained fibers of the long segment of the arcuate fasciculus, and in one third of our subjects, this cluster also contained fibers from the anterior segment of the arcuate fasciculus. The long segment represents the traditional arcuate fasciculus connection between Broca's and Wernicke's area, which is strongly associated with language processing (Catani et al., 2005), whereas the anterior segment connects Broca's region to the inferior parietal region. Catani and colleagues (2005) speculate that a lesion proximal to Broca's region, such as Cluster 1 (overlapping with ROI A), may include both anterior and long segments of the arcuate fasciculus and result in Broca-like aphasia symptoms, such as an impairment in language fluency. Lesions to the long segment of the arcuate fasciculus are often reported to result in conduction aphasia, which is marked by poor speech repetition and phonemic paraphasic errors (Dick & Tremblay, 2012).

Cluster 2 from the whole-brain analysis appears to contain the posterior segment of the arcuate fasciculus. A recent study in adults found that increased fractional anisotropy in the posterior segment of the arcuate fasciculus was related to better reading scores, independent of early schooling experience (Thiebaut de Schotten et al., 2012). This posterior segment appears to connect the temporal lobe with temporo-parietal regions. Remarkably, ROI B (the left temporo-parietal white matter near the arcuate fasciculus), reported in a recent study (Saygin et al., 2013), overlaps with Cluster 2. Saygin and colleagues found a correlation between phonological awareness and the volume and microstructure of the posterior arcuate fasciculus in preliterate children. Our results provide additional evidence that this region may be important for the development of reading skills between kindergarten and Grade 3.

Although our study is novel in several ways, it does have some potential limitations, such as having access to diffusion MRI data only at Time 2. Future studies may limit the sample purely to preliterate children and examine how short-term developmental changes in white matter (e.g., kindergarten through Grade 1) predict long-term reading outcomes (e.g., at Grade 3), which makes these imaging metrics more applicable to reading diagnosis and assessment. Additionally, despite 18 of our participants beginning the study with a family history of reading difficulty, only 1 participant had a formal diagnosis of reading difficulty at Time 2. Finally, we did not include ventral pathways in our ROI analysis because reports of these findings are in native space. For example, studies such as Vandermosten et al. (2012) and Yeatman et al. (2012) were excluded, though these researchers found a significant and unique effect of the ventral pathway (Yeatman et al., 2012).

Nevertheless, our finding that developmental change in left dorsal temporo-parietal white matter predicts reading achievement is unique. We have shown for the first time that this relationship is (a) independent of associated behavioral, demographic, and environmental measures and (b) associated with structural developmental predictors beginning in early kindergarten readers. The study emphasizes the importance of multimodal longitudinal neuroimaging to gain a comprehensive understanding of the brain bases of reading, especially as developmental measures were sensitive predictors of reading outcomes whereas cross-sectional measures were not. The predictive relationships we observed are unlikely to be a widespread means of diagnosis in developmental studies because of cost and time constraints; however, they do point to the critical nature of this time period for reading development. We believe that differences in the rates of structural brain development may have important implications for understanding variations in reading development and the nature and causes of a range of other developmental disorders as well (e.g., Shaw et al., 2007, in relation to ADHD; Yeatman et al., 2012, in relation to reading disorder). Finally, our findings call for a need to investigate what may be influencing the brain during this critical period of reading development with the hope of constructing more targeted and time-appropriate methods of intervention.

Author Contributions

F. Hoeft designed the study. J. M. Black, B. Casto, and M. Tumber contributed to the study design. C. A. Myers, M. Vandermosten, M. Drahos, P. Gimenez, and E. A. Farris analyzed the data. C. A. Myers drafted the manuscript in collaboration with F. Hoeft, C. Hulme, R. Hancock, and R. L. Hendren.

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Declaration of Conflicting Interests

The authors declared that they had no conflicts of interest with respect to their authorship or the publication of this article.

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Supplemental Material

Additional supporting information can be found at http://pss.sagepub.com/content/by/supplemental-data

Note

1. Four out of 28 subjects had more than 1.5 mm of movement. In these participants, we were unable to delineate the arcuate fasciculus. The excessive motion could have impeded our ability to find the arcuate fasciculus, and hence the importance of the arcuate fasciculus may be underestimated by these analyses.

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